

Brief Report on EP4105842A1: Systems and Methods of Situation Aware Edge Analytics Framework for Avionics IoT Gateways

Prepared for Sarada Prasad Moharana

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Abstract

This report summarizes patent EP4105842A1, explains its core technical problem, architecture, relevance to a distributed field intelligence system, and its limitations. The report is intended for a short technical presentation or project literature survey.

1 Basic Identification

- **Publication number:** EP4105842A1
- **Title:** Systems and methods of situation aware edge analytics framework for avionics IoT gateways
- **Technical area:** Industrial/Aerospace Internet of Things, edge analytics, avionics gateways

2 Core Problem

The patent addresses the problem that many IoT and Industrial IoT systems send large amounts of data to the cloud for analysis. This creates latency, bandwidth burden, delayed insights, and poor suitability for time-critical operations. In avionics, analytics has often been restricted to post-flight analysis, which means faults, deviations, or warnings may be discovered too late.

The proposed solution is to add intelligence to the edge platform so that the edge node can adapt to its current operating context and process analytics locally.

3 Main Technical Pipeline

The abstract-level pipeline is:

Server sends rules database → edge device receives sensor/device data → edge device compares data with rules → triggering event is detected → alert is output

The edge node may contain:

- a rules engine;
- edge machine-learning runtime;
- data bus with publish/subscribe topics;
- filtering and transformation modules;

- cloud connector;
- node agent for health/configuration management;
- MQTT/CoAP data ingress and egress.

A key idea is that models or analytics rules can be built or configured centrally, then deployed to the edge node. The edge device processes live data locally and only reports relevant alerts, events, or history updates to the server.

4 Relevance to Field Intelligence

This patent is relevant to the **edge intelligence** layer of the proposed project. A field device cannot rely entirely on cloud processing. It must make local, low-latency decisions when connectivity is weak, delayed, or unavailable.

In a field intelligence system, an analogous design would be:

- if bandwidth is weak, upload only event summaries;
- if visual confidence is low, increase reliance on audio or thermal sensing;
- if battery is low, reduce inference frequency;
- if a high-risk event is detected, send a priority alert immediately;
- if a mission-specific rule is triggered, provide local guidance.

The patent is therefore useful as prior art for *situation-aware edge deployment*, where the device changes behavior depending on context.

5 Limitations

The patent is avionics and Industrial IoT oriented. It is less focused on:

- visual scene understanding;
- audio-visual perception;
- SLAM or spatial mapping;
- general mission-conditioned guidance;
- embodied multi-device field intelligence.

6 Conclusion

EP4105842A1 is important for understanding how analytics can be pushed toward the edge. It supports the design idea that field intelligence devices should not merely stream raw data to a cloud server; they should locally evaluate rules, models, context, health, and connectivity, and should report only useful alerts or outputs when appropriate.

Reference

Google Patents, “EP4105842A1 - Systems and Methods of Situation Aware Edge Analytics Framework for Avionics IoT Gateways.” <https://patents.google.com/patent/EP4105842A1/en>